

Proposal

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Functional Requirements

Overview

Create an app based on adopting and caring for virtual pets with an online database.

Screens

1. **Adoption Screen:** Display all pets available for adoption in a list. Each pet's profile will include:

Image: A picture

Details: Name and age

Adopt Button: By pressing this it will update its status to adopted.

2. **My Pets Screen:** Show a list of the user's adopted pets. Each pet card will include the pet's image, name, and current hunger/happiness level. By clicking on a pet card, the user will be brought to the pet care screen this pet.
3. **Pet Care Screen:** Screen showing the pet. From here the user can give it food, interact with it or clean it. After a set number of seconds each (e.g. hunger) level will gradually decrease.

Database

When the user is on the adoption screen they can choose to adopt a pet. The pets statistics are updated in the database. The pet's hunger and happiness levels change often, which means there will be continuous database interaction. Users can also release the pet which will remove it from their database. The database will track each user and what pets they have along with the pets data.

Navigation

Ensure smooth and quick navigation between screens (Adoption, My Pets, Pet Care, and Profile), making the app engaging and user-friendly. This will be achieved through buttons and a navigation bar.

UI State Management

Each screen will use ViewModel to manage the UI state. This avoids unnecessary queries and improves performance.

Service

In the pet care screen when the user clicks the heart icon and then brings their hand (or face) close to the device the proximity sensor will detect the movement, and the pet will start to jump. Each time it happens the pet's happiness level will increase.

Custom theme

The app will have a fun and playful aesthetic by featuring a theme with soft pastel colors and rounded edges. This will make the app feel inviting to users.

Material Design guidelines

The app follows the material design principles as it has a consistent layout, floating action buttons, and clear fonts. Each action adheres to the design standards, ensuring a familiar and a user-friendly experience.

Visually appealing

I have designed each screen with attractive visuals which include, pet images, data and buttons. The colour complement each other while the text on the screen is easy to read.

User experience

With user friendly navigation, seamless updates and a colourful theme the app prioritises user experience. It is possible to adopt and interact with pets quickly and statistics are easy to access ensuring a fun user experience. Notifications could be implemented to alert a user if another pet needs something while they are at a different screen.

Battery Efficiency

The app is memory and battery-efficient as it is using ViewModel to cache data. This will minimise data and memory usage. The app uses the proximity sensor only when the user presses the heart button in the pet care screen, intentionally activating the

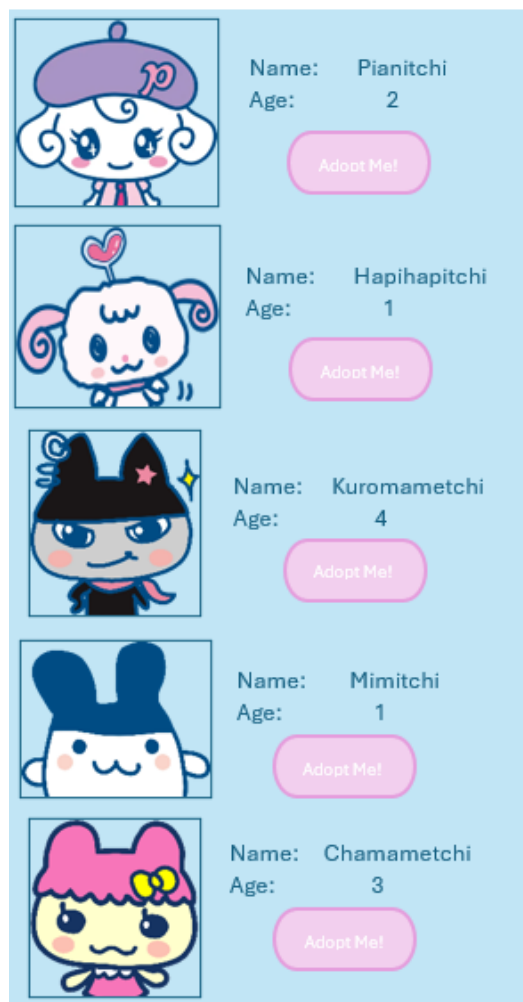
sensor. This will conserve battery as it will not continuously use the sensor, instead only activating it for a brief period before it is disabled again, therefore preventing the battery from being drained excessively.

Creativity

I believe this is a very creative way to implement what is essentially a booking system. Since you can adopt (book) a pet. The pets statistics are updated in the database and users can release the pet (cancel). Along with the visually appealing design of the app layout and screens.

Screen Mockups

Adoption Screen



My Pets Screen

	Name: Pianitchi Age: 2 Happiness: 6/10 Hunger: 5/10
	Name: Hapihapitchi Age: 1 Happiness: 8/10 Hunger: 7/10
	Name: Mametchi Age: 4 Happiness: 10/10 Hunger: 10/10
	Name: Melodytchi Age: 2 Happiness: 9/10 Hunger: 9/10
	Name: Lovelitchi Age: 1 Happiness: 9/10 Hunger: 8/10

Pet Care Screen

